

Air Quality Index Predictions with a Hybrid Forecasting Model: Combining Series Decomposition and Deep Learning Techniques

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Abstract—As China's industrialization process and people's living standards continue to improve, the importance placed on air quality has gradually increased. Precise prediction of the Air Quality Index (AQI) is vital for proactively mitigating the risks posed by air pollution. However, air quality levels can fluctuate significantly over a certain period. Even the interval models used in recent research face the challenge of not being able to retain essential information regarding air quality status. The prediction of the Air Quality Index (AQI) is crucial for controlling air pollution. Furthermore, while pursuing longer prediction times, the accuracy of the prediction must also be ensured. This paper proposes a model based on the fusion of EEMD-CEEMDAN and LSTM. EEMD-CEEMDAN is used to decompose the raw data to reduce its complexity, improve the accuracy and stability of data analysis, and then LSTM is used to predict the AQI. The experiments show that the model proposed in this paper demonstrates excellent experimental results on the real-world collected weather quality dataset.

Keywords—AQI prediction, EEMD, CEEMDAN, LSTM, Deep Learning

I. INTRODUCTION

Nowadays, China's economy and industrialization have been rapidly growing, expanding urban populations and industrial scale. With the increased use of fossil fuels such as coal and oil, the levels of gaseous and solid pollutants in the air have also risen accordingly. Leading to air pollution getting worse and worse[1]. The air pollution in some areas has become so severe that it affects people's daily lives. Air pollution has attracted extensive attention from all countries. Scientific research has found that exposure to severely polluted air can cause many serious health problems in the human body, especially in vital parts such as the lungs and cardiovascular system[2,3,4,5]. If the

exposure time is too long, it may even cause cancer, damage the immune system, and even damage the reproductive system. Air pollution not only harms the ecological environment but also severely restricts economic development[6,7]. The Air Quality Index (AQI) directly reflects air quality, with better air quality corresponding to a lower AQI[8,9]. Accurate predictions of AQI can enhance the effectiveness of air pollution prevention and control. However, due to the dynamic and nonlinear nature of air distribution in the environment, analyzing and predicting AQI have proven particularly challenging[10,11,12]. As a result, researchers have devoted significant attention to exploring air data's non-stationarity and nonlinearity to improve AQI predictions' accuracy.

The model presented in this article aims to forecast air quality data, specifically the sequence of AQI data. Initially, the EEMD[13] method is employed to decompose all air components, except for AQI data, into various modes, resulting in multiple different features for each. Then, the AQI data is decomposed by the CEEMDAN[14] method to obtain the various ways of AQI sequence data. Finally, the mode components with higher correlation with AQI in step one are selected and sent to the LSTM[15] recurrent neural network as features along with the mode components obtained through the decomposition of AQI, in order to obtain the prediction results.

II. RELATED WORK

Currently, three primary categories of AQI forecast models exist: statistical forecast models, machine learning forecast models, and hybrid forecast models. Within the statistical model's group, there are several subtypes, such as autoregressive integrated moving average models (ARIMA)[16], multivariate linear regression models (MLR)[17,18], and grey models

(GM)[19,20]. However, these models have poor stability in predicting nonlinear data. As artificial intelligence continues to evolve, machine learning models have been widely used in air quality prediction, including artificial neural networks (ANN)[21], convolutional neural networks (CNN), and support vector machines (SVM) [23].

As a subfield of machine learning, deep learning is a powerful machine learning method that is more suitable for big data. Deep learning has achieved impressive results in time series prediction. Currently, Two primary deep learning models used for time series prediction are based on deep networks, specifically RNN (Recurrent Neural Network) and LSTM (Long Short-Term Memory)[15]. These models are designed to capture and model time information effectively.

Many experiments have shown that many hybrid models have been used for time series prediction tasks and have achieved good prediction results [24,25]. Zhu et al. employed empirical mode decomposition (EMD) and ensemble empirical mode decomposition (EEMD) to decompose the AQI sequence signal, which resulted in enhanced prediction performance after the decomposition process [26,27]. EMD may encounter mode mixing issues in many experiments, and its lack of rigorous mathematical proof could also become a concern in some instances [29].EEMD is an enhanced method built on top of EMD, which offers an effective solution for the mode mixing issues encountered in EMD. Torres et al. [30] introduced the adaptive noise complete ensemble empirical mode decomposition with adaptive noise (CEEMDAN), which builds on EMD and leverages techniques such as adding Gaussian noise and multiple superimpositions and averaging to reduce noise in EEMD. This method effectively addresses the mode mixing problem while improving upon EMD. Yang et al. [28] utilized complete ensemble empirical mode decomposition with adaptive noise (CEEMDAN) to decompose the AQI sequence data. Afterward, they employed a neural network to forecast a range of decomposed components before merging the predicted outcomes for each element to acquire the final prediction result. Experimental results indicate that hybrid prediction models offer superior performance in terms of prediction accuracy compared to direct prediction models.

III. PROPOSED METHOD

The model proposed in this article achieves precise prediction of AQI by integrating EEMD, CEEMDAN, and LSTM, which can enhance the efficiency of air pollution prevention and control efforts. The following steps are involved:

- Preprocess the air quality data collected by sensors, fill missing values with averages, and arrange all data chronologically.
- Compose each air quality data component into its modal components using the EEMD method.
- Calculate the correlation between each modal component obtained in step 2 and the AQI sequence data, and select the modal features whose correlation coefficient is more significant than 0.35.

- Decompose the AQI sequence data using the CEEMDAN method to obtain the modal components of AQI.
- Utilize the modal components acquired from steps 3 and 4 as features to train LSTM and obtain the predicted results as output. The framework of the model is shown in Figure. 1.

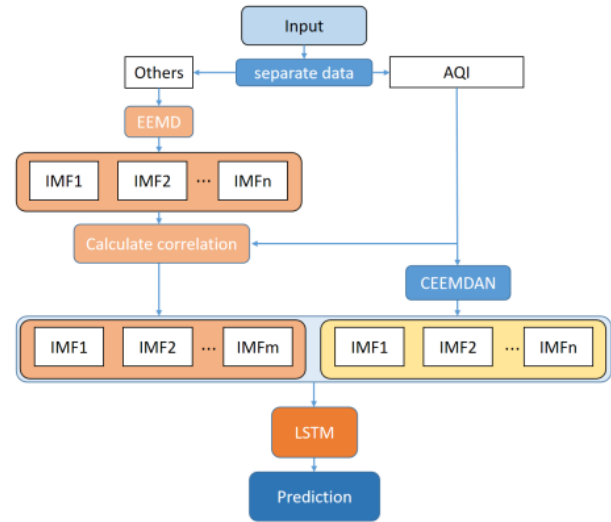


Fig. 1. The framework of the model

A. Data Preprocessing

Data preprocessing is of utmost importance in data analysis, as the data's content directly impacts the ultimate prediction's reliability and precision. During data preprocessing, we check the completeness of the data, delete erroneous data, and use mean values to fill in deleted and missing data. Although the data already meets the format required for model training, To enhance the efficiency and efficacy of model computations and avoid the issue of overfitting, this article adopts the method of maximum and minimum value normalization to standardize the preprocessed data.

B. Prediction Model

The prediction model is trained using the preprocessed dataset as input, which learns the time correlation between AQI data and other related data to achieve AQI sequence data prediction. The prediction model consists of a combination of EEMD-CEEMDAN-LSTM, where EEMD and CEEMDAN are used for signal decomposition of time series data to reduce the difficulty of data analysis. The purpose of employing LSTM is to understand the temporal correlation within time series data.

- EEMD is the most common improvement of EMD. Its principle is to use the characteristic of Gaussian white noise with mean 0 by artificially adding noise multiple times in the process of signal decomposition to mask the noise of the signal itself to obtain more accurate upper and lower envelope lines. Simultaneously, the decomposition outcomes are averaged, and with each iteration of the averaging process, the impact of noise on the decomposition is diminished. The EEMD process diagram is shown in Figure. 2.

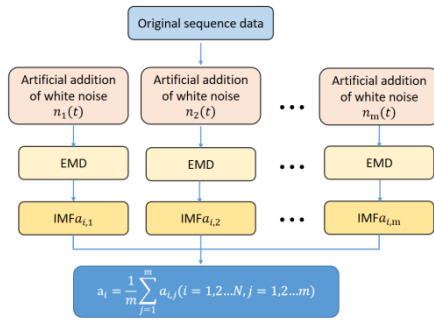


Fig. 2. The EEMD process diagram

- CEEMDAN: The decomposition method that is improved based on EMD also draws on the idea of adding Gaussian white noise in EEMD and superimposing and averaging each mode component multiple times to counteract the noise of the signal itself. The CEEMDAN process diagram is shown in Figure. 3.

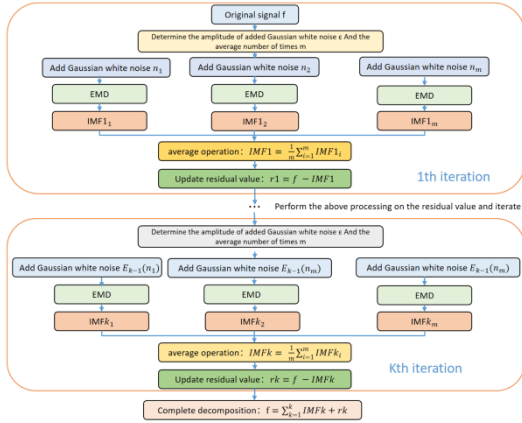


Fig. 3 The CEEMDAN process diagram

- LSTM: LSTM (Long Short-Term Memory) refers to a distinctive class of recurrent neural networks, as shown in Figure. 4, mainly designed to solve the problems of gradient vanishing and exploding in the training process of regular recurrent neural networks when dealing with long sequence data. To put it simply, in contrast to conventional RNNs, LSTM excels in processing lengthier sequences, making it a sought-after choice for time series prediction endeavors.

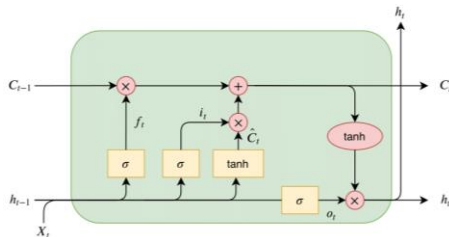


Fig. 4 The schematic diagram of the LSTM structure

From Figure 4, it can be seen that LSTM neurons are mainly composed of four parts: forget gate f_t , input gate i_t , state gate c_t , and output gate o_t . The operation process is as follows:

1) By using the data of LSTM neurons, the data is first selectively forgotten through the forget gate f_t :

$$f_t = \sigma(W_{hf}h_{t-1} + W_{xf}x_t + b_f) \quad (1)$$

2) Then enter the input gate i_t :

$$i_t = \sigma(W_{hi}h_{t-1} + W_{xi}x_t + b_i) \quad (2)$$

3) Calculate the state value of the data after passing through the forget gate and input gate:

$$c_t = f_t c_{t-1} + i_t \tilde{c}_t \quad (3)$$

$$\tilde{c}_t = \tanh(W_{xc}x_t + W_{hc}h_{t-1} + b_c) \quad (4)$$

4) Obtain the final output result of LSTM neurons through the output gate:

$$o_t = \sigma(W_{ho}h_{t-1} + W_{xo}x_t + b_o) \quad (5)$$

$$h_t = o_t \tanh(c_t) \quad (6)$$

IV. EXPERIMENTAL RESULTS AND DISCUSSION

This section will introduce the dataset used in this experiment, the experimental settings of the model, and the details of the experimental results obtained, and will be described in detail in the following chapters.

A. Data Set

This paper uses the accumulated five-year air quality data from Guangzhou City from 2017 to 2021 as the dataset, which can be downloaded from the China National Environmental Monitoring Centre (<http://www.cnemc.cn/>) and used as experimental data. The data collects air quality data every other day from January 1, 2017 to August 14, 2021, including the concentration of various air quality components such as AQI, CO, NO₂, O₃, PM₁₀, PM_{2.5}, etc. A total of 1681 observation data were selected as the experimental data, which were then divided into a training set, validation set, and test set in a ratio of 8:1:1. Figure. 5 shows the visual display of the changes over time of various components in the dataset used in this experiment.

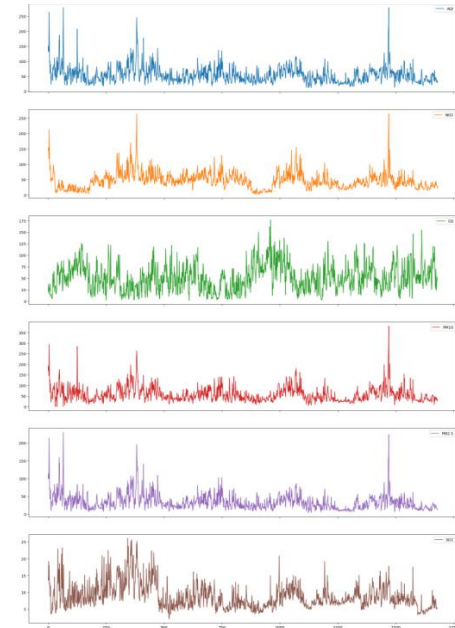


Fig. 5. Data display

B. Evaluation Indicators

To quantify the performance of the models, this paper uses four metrics: MAE, MSE, MAPE, and R^2 to assess the performance of the models. Mean Absolute Error (MAE) is employed to gauge the degree of proximity between the predicted outcomes and the actual dataset; with the lower the value, the more favorable the fitting outcome. Mean Squared Error (MSE) is a metric that indicates the average of the squared deviations between the fitted data and the initial data related to the sample points, the lower the value, the more satisfactory the fitting outcome. The Mean Absolute Percentage Error (MAPE) is a relative measurement of error that employs absolute values to circumvent positive and negative errors from canceling each other out. The more diminutive the MAPE, the more apt the model fitting consequences. R^2 (R Squared) corresponds to the coefficient of determination, denoting the level of model fit. The range of values fluctuates within $[0,1]$, where a higher value closer to 1 suggests more favorable results.

C. Results

To assess the efficacy of the proposed model, the study compared the performance of LSTM and Bi-LSTM models on the identical dataset, as outlined in Table I. The results show that the proposed model outperforms the LSTM, Bi-LSTM, and VMD-LSTM models across all parameters.

TABLE I. OVERALL RESULTS COMPARISON WITH BASELINE MODELS

Models	MAE	MSE	MAPE	R^2
LSTM	13.49	461.98	22.62	0.29
Bi-LSTM	13.61	488.96	23.48	0.20
VMD-LSTM	11	306.94	19.07	0.55
EEMD-CEEMDAN-LSTM	7.05	128.65	11.69	0.84

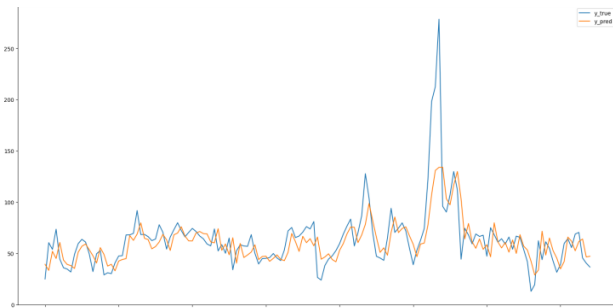


Fig. 6. The performance of LSTM

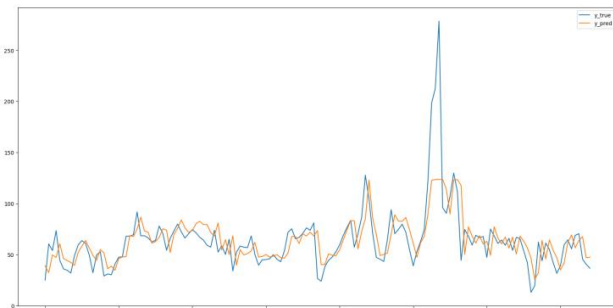


Fig. 7. The performance of Bi-LSTM

The performance of LSTM, Bi-LSTM, VMD-LSTM, and EEMD-CEEMDAN-LSTM on the test set is shown in the Figures. 6, 7, 8, and 9, respectively.

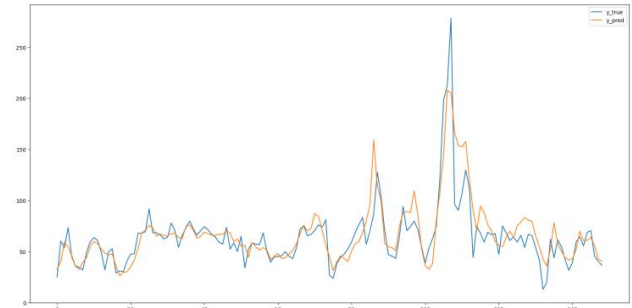


Fig. 8. The performance of VMD-LSTM

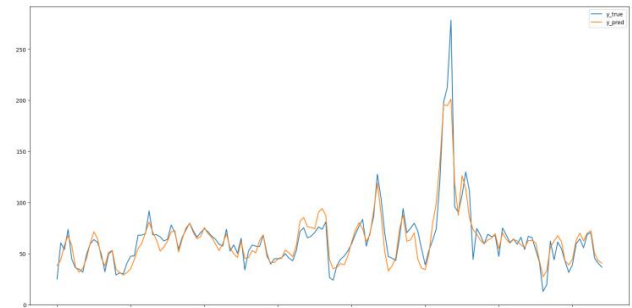


Fig. 9. The performance of EEMD-CEEMDAN-LSTM

This study used the Ensemble Empirical Mode Decomposition (EEMD) for decomposing complex signals into intrinsic mode functions. This approach has several benefits, including:

- **Adaptability:** EEMD is highly adaptable and can be used for various types of signals, including non-stationary, nonlinear, and non-periodic signals.
- **Noise reduction:** EEMD can effectively filter out noise and extract meaningful signals from noisy data, which can improve the accuracy of data analysis and modeling.
- **No mode mixing:** Unlike other decomposition methods, EEMD does not suffer from mode mixing, where different modes of a signal get mixed together, making it difficult to analyze the individual components.
- **Robustness:** EEMD is a robust technique that can handle data with missing values, outliers, and other anomalies, making it useful in real-world applications.
- **Ensemble averaging:** EEMD involves averaging multiple decompositions to obtain a more reliable decomposition result, which can improve the accuracy and stability of the analysis.

V. CONCLUSION

To refine the precision of AQI prognostication, this paper proposes a hybrid prediction model for the air quality index that harnesses EEMD-CEEMDAN-LSTM, which combines the signal decomposition with machine learning. Compared with

traditional prediction models, the proposed model in this paper has greatly improved on every evaluation index in experiments on the same dataset. The presented approach was tested using data collected from air quality stations in Guangzhou China. The experiments demonstrated that this method outperforms comparable models in terms of accuracy, and the analysis of its application suggests that it has potential for use in AQI online forecasting systems. In summary, this paradigm offers a promising alternative for forecasting AQI time series.

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